

Thermal Comfort Prediction Model for Indoor Spaces

Group # 19

Members: Arshman Tariq, Jamiual Topu, Md Nuruzzaman Faruk
Muhammad Faizan, Tanveer Naganandhini, Somalinga Mohanlal

Course: Internet of Things (IoT)
15 December 2025



Motivation & Overview

Why This Project?

Problem

- Indoor thermal discomfort in Nordic climates
- Extreme heating systems + humidity fluctuations
- Physiological impacts: throat irritation, dry skin, respiratory issues

Solution

- Real-time IoT monitoring system
 - Continuous environmental tracking
 - Enable proactive corrective measures (ventilation, heating adjustments)
- 

IoT Architecture Overview

1. Sensing Layer

Pico W + BME280 sensor + LED/LCD feedback

2. Networking Layer

MQTT (HiveMQ Cloud, SSL/TLS, port 8883)

3. Data Management Layer

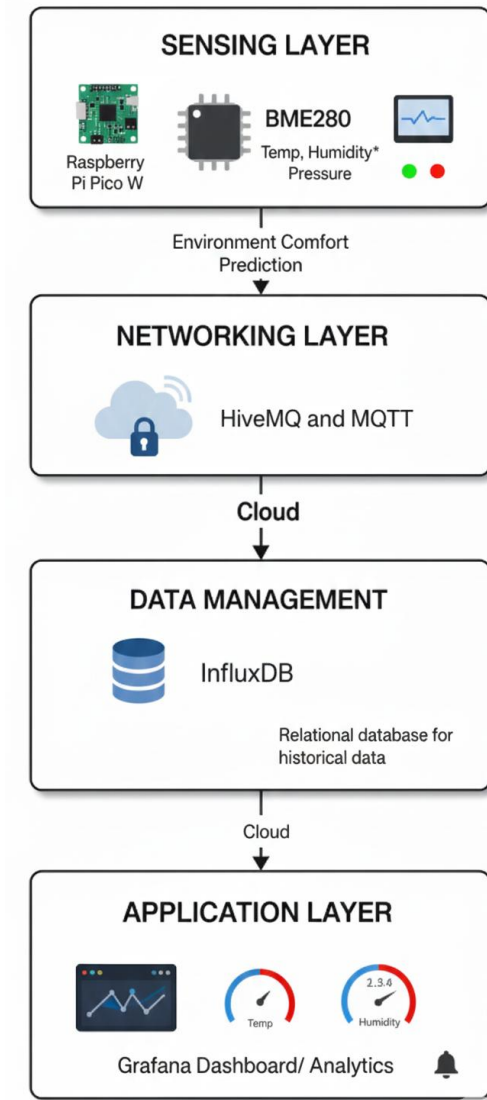
Node-RED middleware → InfluxDB (time-series storage)

4. Application Layer

Grafana dashboards + real-time alerts

5. Security Layer

End-to-end encryption + credential authentication



Mandatory and Optional Functionalities

What MUST be Implemented

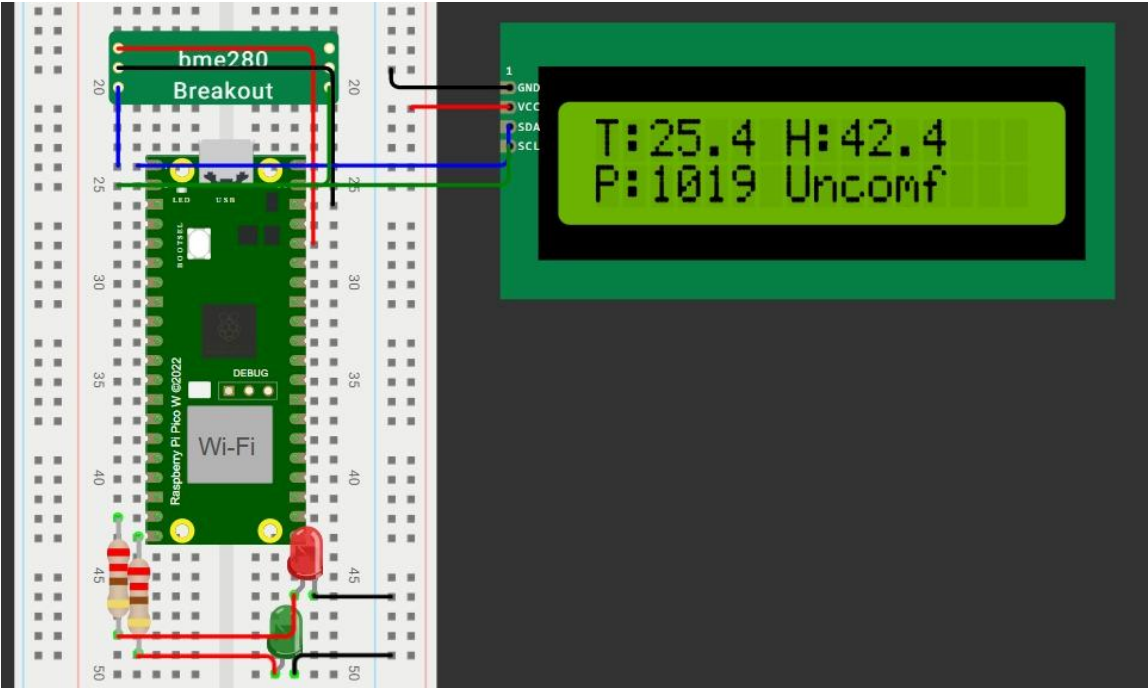
- ✓ Sensor Data Collection
 - BME280 sensor via I2C bus
 - Real-time: Temperature, Humidity, Atmospheric Pressure
- ✓ Data Processing
 - Rule-based and ML-based comfort classification on Pico W
 - Thresholds: 20-25°C + 30-60% humidity = Comfortable
- ✓ Output Control
 - 16x2 LCD display (real-time values + status)
 - Bi-chromatic LEDs
 - Green LED = Comfortable conditions
 - Red LED = Uncomfortable conditions

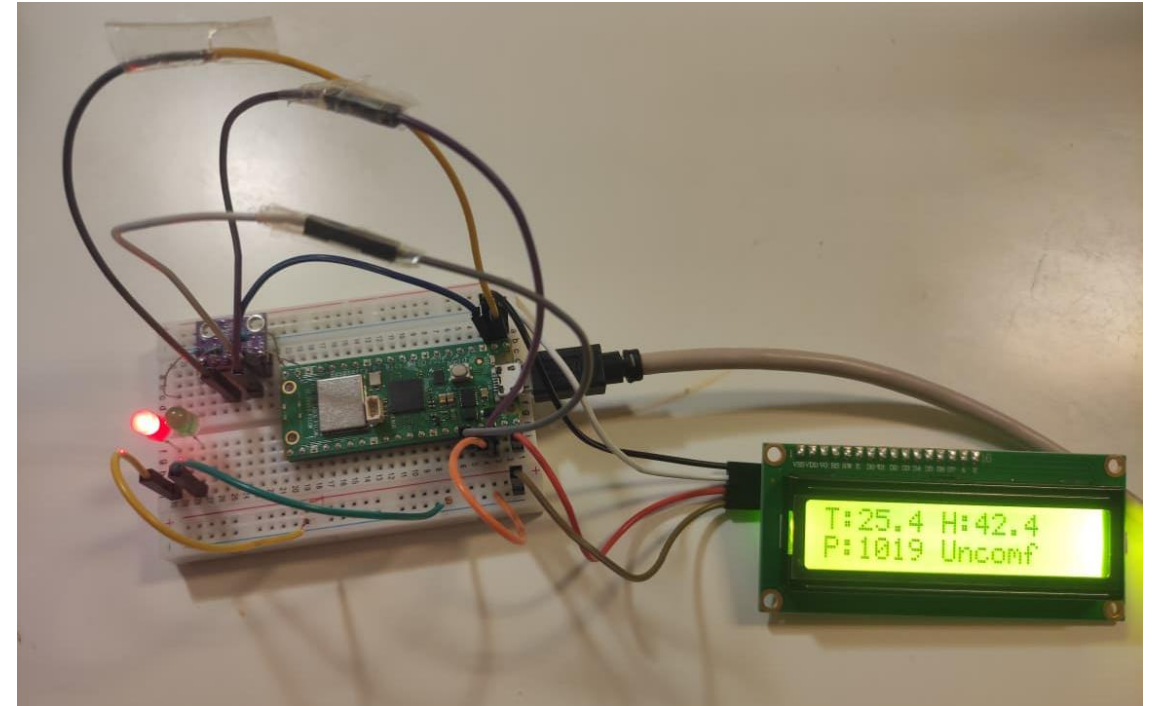
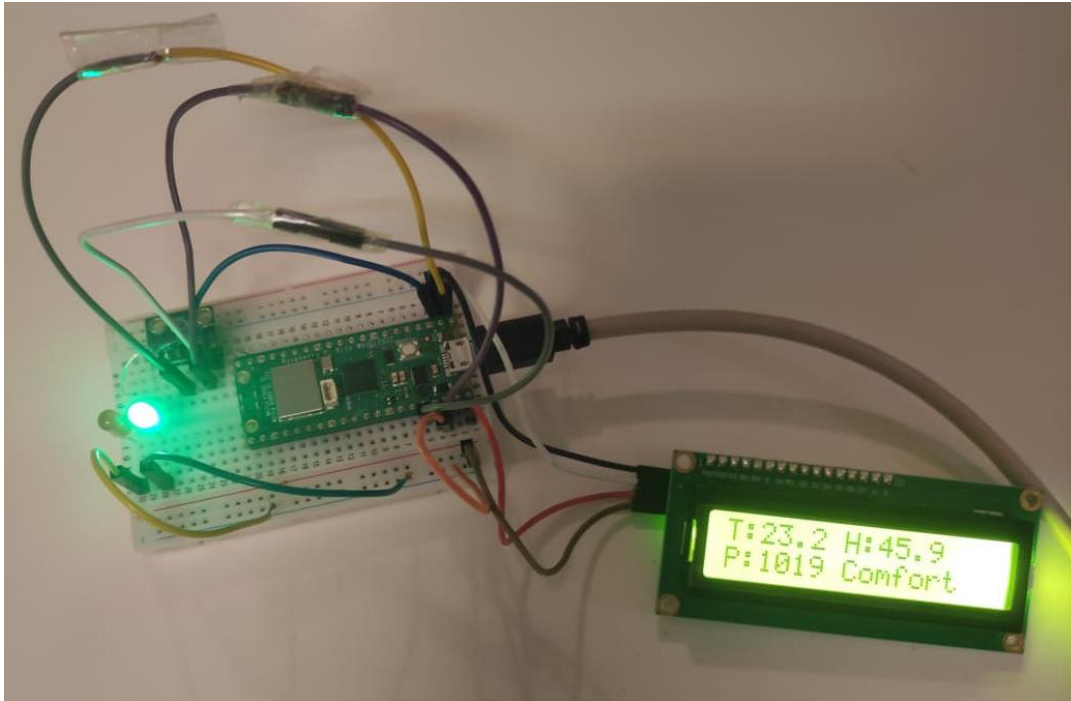
3 out of 5 Optional Functionalities IMPLEMENTED

1.  Wireless Connectivity
MQTT protocol (lightweight, publish-subscribe)
2.  Cloud Integration
InfluxDB + Grafana dashboards + email alerts
3.  Edge Machine Learning
Random Forest classifier (10 trees, max depth 5)

Hardware Components & Circuit Diagram

Component	Specs	Purpose
Raspberry Pi Pico W	2MB flash, 264KB RAM Onboard Wi-Fi	Edge computing
BME280 Sensor	I2C (addr: 0x77)	Temperature Humidity Pressure
16x2 LCD Display	I2C PCF8574 (addr: 0x27)	Display feedback
LEDs	GPIO 14 (Red) GPIO 15 (Green)	Visual indicators
Resistors	220 Ω	Current limiting





Hardware Output

Red LED Actuation (Triggers when temperature or humidity out of range)

Green LED Actuation (Confirms comfortable environmental conditions)

LCD Display (Shows real-time sensor values, Humidity, Pressure, Temperature and Comfort status)

Model Classification Logic

Rule-Based Model (Primary - Lightweight)

Decision Logic:

IF temperature $\in [20, 25]^{\circ}\text{C}$ AND humidity $\in [30, 60]\%$

→ Comfortable (Green LED ON)

ELSE

→ Uncomfortable (Red LED ON)

ML-based Model (Extended Work)

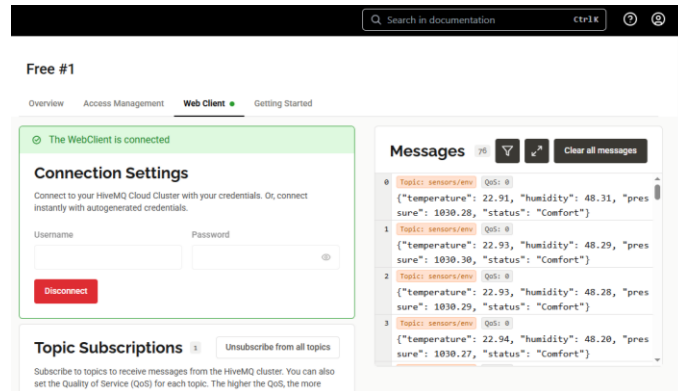
Algorithm: Random Forest

- 10 decision trees, max depth 5
- Training data: 200 labeled environmental samples
- Test data: 44 samples (hold-out validation)

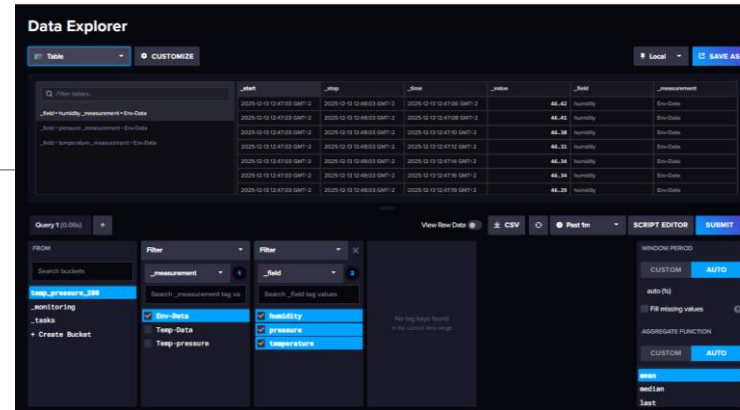
Feature Importance:

- Humidity: 36.3% (most influential)
- Temperature: 35.3%
- Pressure: 28.4%

Software Outputs



HiveMQ Interface



InfluxDB Interface

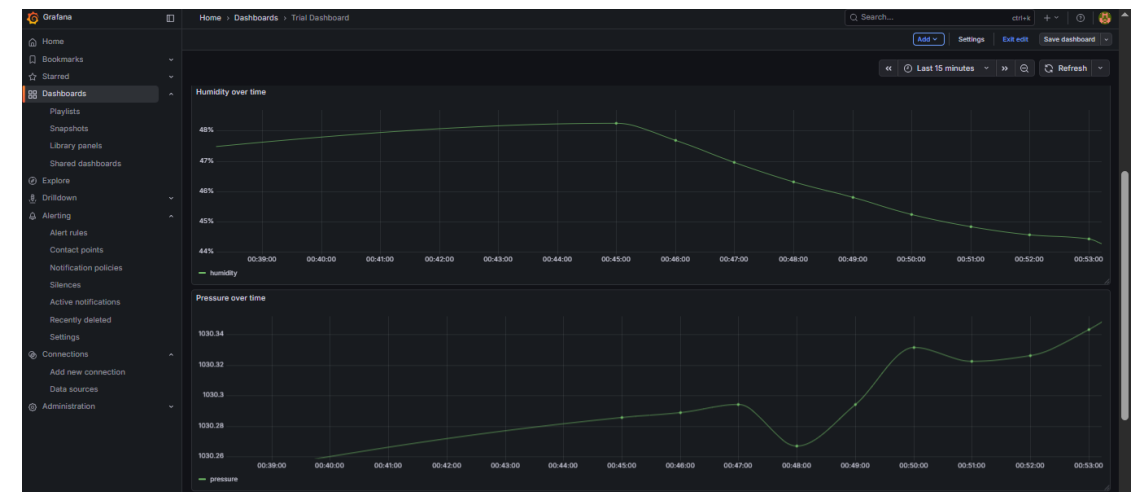
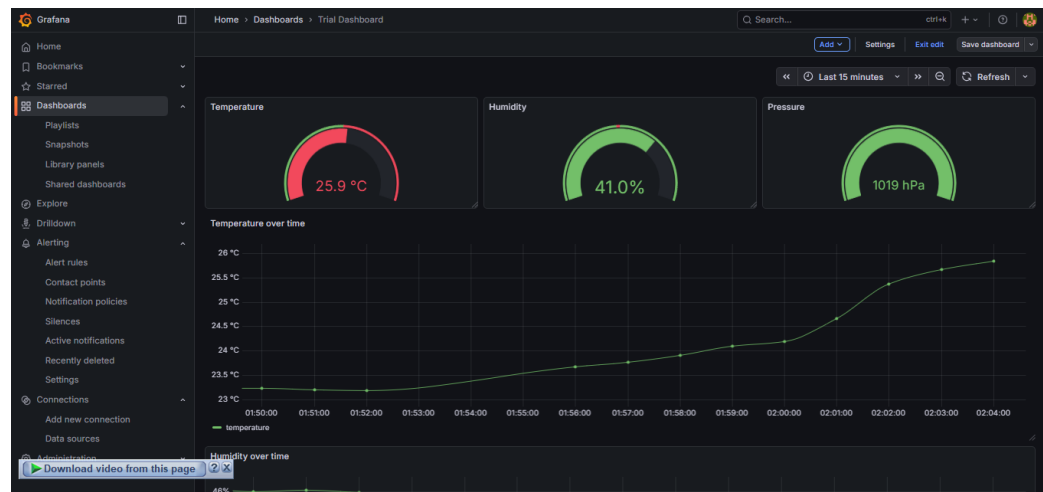


Demo folder_latest › DatasourceError

Hi, Your Environment is Uncomfortable.

© 2025 Grafana Labs. Sent by [Grafana v12.3.0](#).

Grafana Email Alerts



Grafana Interface (Pressure, Humidity, Temperature Gauges and Historical Trends)

System Performance Summary		
Metric	Result	Status
MQTT Delivery Rate	241/241 messages (100%)	✓ Perfect
Message Latency	95-320 ms (avg: 180 ms)	✓ Acceptable
Throughput (2s sampling)	0.52 msgs/sec	✓ Stable
Throughput (0.5s sampling)	0.67 msgs/sec	✓ Robust
Payload Impact	0.46-0.49 msgs/sec (76-143 bytes)	✓ Minimal overhead
ML Accuracy	97.7% (44 test samples)	✓ Excellent
Continuous Operation	1+ hours tested	✓ Reliable

Evaluation Metrics & Results

Future Work/ Improvements:

- Automated actuators (smart window opening, HVAC control)
- Deep sleep modes for battery-powered deployments
- Multi-class comfort categories (very cold → very hot)
- Multi-node deployment at the building level with spatial analytics
- Resilience to network outages (local caching)



GITHUB REPOSITORY

[HTTPS://GITHUB.COM/FAIZANTANWIR/IOT_PROJECT_GROUP_19.GIT](https://github.com/faizantanwir/IOT_PROJECT_GROUP_19.GIT)